(iv) A ray falling on the pole will get reflected as per law of reflection keeping principal axis as the normal. (Figure 6.10(d))

6.2.4 Cartesian sign convention

While tracing the image, we would normally come across the object distance u, the image distance v, the object height h, the image height h', the focal length f and the radius of curvature R. A system of signs for these quantities must be followed so that the relations connecting them are consistent in all types of physical situations. We shall follow the Cartesian sign convention which is now widely used. It is given below and also shown in Figure 6.11.

(i) The Incident light is taken as if it is travelling from left to right (i.e. object on the left of mirror).
(ii) All the distances are measured from the pole of the mirror (pole is taken as origin).
(iii) The distances measured to the right of pole along the principal axis are taken as positive.
(iv) The distances measured to the left of pole along the principal axis are taken as negative.
(v) Heights measured upwards perpendicular to the principal axis are taken as positive.
(vi) Heights measured downwards perpendicular to the principal axis, are taken as negative.

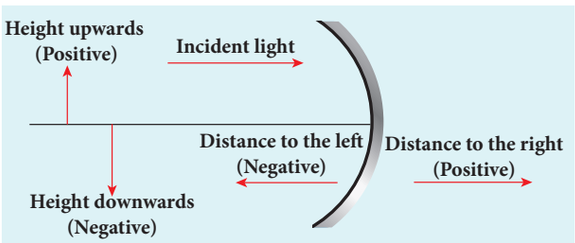


Figure 6.11 Cartesian sign convention

6.2.5 Mirror equation

The mirror equation establishes a relation among object distance u, image distance v and focal length f for a spherical mirror.

An object AB is considered on the principal axis of a concave mirror beyond the centre of curvature C. The image formation is shown in the Figure 6.12. Let us consider three paraxial rays from point B on the object. The first paraxial ray BD travells parallel to the principal axis. It is incident on the concave mirror at D, close to the pole P. It is reflected back through the focus F. The second paraxial ray BP is incident at the pole P. It is reflected along PB'. The third paraxial ray BC passing through centre of curvature C, falls normally on the mirror at E. It is reflected back along the same path. The three reflected rays intersect at the point B'. A perpendicular drawn as $A'B'$ to the principal axis gives the real, inverted image.

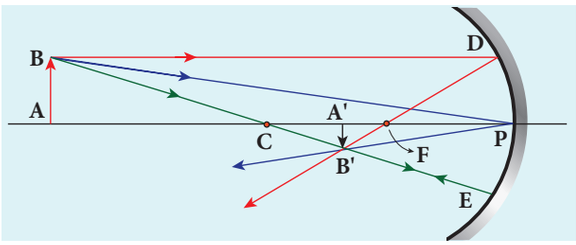


Figure 6.12 Mirror equation

As per law of reflection, the angle of incidence $\angle BPA$ is equal to the angle of reflection $\angle B'PA'$.

The triangles ΔBPA and $\Delta B'PA'$ are similar. Thus, from the rule of similar triangles,

$$\frac{A'B'}{AB} = \frac{PA'}{PA} \qquad (6.5)$$

The other set of similar triangles are, ΔDPF and $\Delta B'A'F$. (PD is almost a straight vertical line)

UNIT-6(XII-Physics_Vol-2)_new.indd 8 21-07-2020 16:46:33